
Binary Tree — Complete Guide with Intuition, Visuals, and C++ Implementation

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1) Build Intuition First: What is a Binary Tree?

Think of a company hierarchy:

- CEO at the top
- each manager may have up to **two direct reports**
- each report may also have up to two reports
- this forms a **branching structure**

That is the core idea of a **binary tree**.

Definition

A **binary tree** is a hierarchical data structure where:

- each node stores some data
- each node can have **at most two children**
    - **left child**
    - **right child**

Real-world intuition

Imagine folders:

A folder may split into:

- left subfolder
- right subfolder

Imagine decisions:

“Is user authenticated?”

- yes → go left
- no → go right

Imagine tournament brackets:

Each round combines two participants under one parent node.

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2) Binary Tree Visualization

Here is a simple binary tree:

A

/

B C

/

D E F

Terms

- A = **root**
- B and C = children of A
- D, E are children of B
- F is right child of C
- D, E, F = **leaf nodes** (no children)
- B, C = **internal nodes**
- Every node with its descendants forms a **subtree**

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3) Key Terminology

3.1 Root

The topmost node.

10 ← root

/

5 20

``

3.2 Parent / Child

If node A points to B, then:

- A is parent
- B is child

3.3 Siblings

Nodes with the same parent.

A

/

B C

B and C are siblings.

3.4 Leaf node

A node with no children.

A

/

B C

/

D

Leaves: B, D

3.5 Height

The number of edges in the longest path from the node to a leaf.

Example:

A

/

B C

/

D

- Height of D = 0
- Height of B = 1
- Height of A = 2

3.6 Depth

Distance from root to a node.

- Depth of root = 0
- Depth of its children = 1
- and so on

3.7 Level

All nodes at the same depth belong to the same level.

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4) Types of Binary Trees

Understanding these helps a lot.

4.1 Full Binary Tree

Each node has either:

- 0 children, or
- exactly 2 children

Example:

```
    1
   / </span>
  2   3
 / </span>
4   5
```

Node 2 has 2 children, node 3 has 0 → valid full tree.

4.2 Complete Binary Tree

All levels are completely filled except possibly the last, and the last level is filled from **left to right**.

```
    1
   / </span>
  2   3
```

/   /

4 5 6

```

This is complete.

---

**4.3 Perfect Binary Tree**

All internal nodes have 2 children and all leaves are at the same level.

```
      1

     /

    2   3

   / \ /

  4 5 6 7
```

---

**4.4 Balanced Binary Tree**

The height is kept relatively small, so operations remain efficient.

Examples:

- AVL tree
- Red-Black tree

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**4.5 Degenerate / Skewed Tree**

Looks like a linked list.

```
  1

    2

      3

        4
```
```

This is bad for performance.

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5) How is a Binary Tree Represented in C++?

Each node stores:

- data
- pointer to left child
- pointer to right child

Node structure

```
struct Node {
  int data;
  Node* left;
  Node* right;

  Node(int value) : data(value), left(nullptr), right(nullptr) {}
};
```

---

6) Core Operations on Binary Tree

For a **general binary tree**, common operations are:

1. Create node
2. Insert node
3. Delete node
4. Search node
5. Traversals
    - Preorder
    - Inorder
    - Postorder
    - Level order
6. Height
7. Count nodes

8. Count leaf nodes
9. Mirror / invert tree

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7) Important Note: Binary Tree vs Binary Search Tree

A **binary tree** does **not** enforce ordering.

Example:

```
  10
 /  </span>
99  2
```

This is still a binary tree.

But in a **Binary Search Tree (BST)**:

- left subtree values < root
- right subtree values > root

Since you asked for **binary tree**, I'll focus on the **general binary tree**, not BST.

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8) Operation 1 — Insertion in a Binary Tree

In a general binary tree, insertion is not naturally ordered.
A common strategy is **level-order insertion**:

- insert the new node at the **first empty position**
- helps keep tree relatively compact

8.1 Insertion Visualization

Suppose we have:

```
    1
   / </span>
  2   3
 /
4
```

Insert 5.

We scan level by level:

- node 1 → both filled
- node 2 → right is empty → insert there

Result:

```
    1
   / </span>
  2   3
 / </span>
4   5
```

8.2 High-Level Architecture for Insertion

Idea

Use a **queue** for level-order traversal:

1. If tree is empty → new node becomes root
2. Push root to queue
3. Pop nodes one by one
4. If left child empty → insert there
5. Else push left child
6. If right child empty → insert there
7. Else push right child

Why queue?

Because queue helps us explore nodes **level by level**.

8.3 Time Complexity

- **O(n)** in worst case
- Space: **O(n)** for queue

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9) Operation 2 — Deletion in a Binary Tree

Deletion in a generic binary tree is different from BST deletion.

Common strategy

To delete a node:

1. Find the target node
2. Find the **deepest rightmost node**
3. Copy deepest node's value into target node
4. Delete the deepest node

This works well for a general binary tree.

9.1 Deletion Visualization

Initial tree:

```
      1
     / </span>
    2   3
   /    </span>
  4  5  6
```

Delete node 2.

Step 1: Find node with value 2

Step 2: Find deepest rightmost node → 6

Step 3: Replace 2 with 6

Intermediate:

```
      1
     / </span>
    6   3
   /    </span>
  4  5  6
```

Step 4: Remove original deepest node

Final:

```
      1
     / </span>
```

6 3

/

4 5

9.2 High-Level Architecture for Deletion

Algorithm

1. If tree empty → return
2. If root is the only node and matches target → delete root
3. Perform level-order traversal:
    - keep track of:
        - targetNode (node to delete)
        - deepestNode
        - parent of deepest node
4. If target found:
    - replace target value with deepest value
    - remove deepest node from its parent

9.3 Time Complexity

- **O(n)** time
- **O(n)** space

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10) Operation 3 — Searching in a Binary Tree

Because the tree is not ordered, you may need to inspect many nodes.

10.1 Search Visualization

Search for 5 in:

1

/

2 3

/

4 5 6

Possible level-order search path:

1 → 2 → 3 → 4 → 5

Found at node 5.

10.2 High-Level Architecture for Search

Option 1: Recursive DFS

- check current node
- search left subtree
- search right subtree

Option 2: Iterative BFS

- use queue
- scan level by level

10.3 Time Complexity

- Worst case: **O(n)**

---

11) Operation 4 — Tree Traversals

Traversal means visiting all nodes in some order.

These are very important.

11.1 Depth-First Traversals

Given tree:

A

/

B C

/

D E F

```

---

**(a) Preorder: Root → Left → Right**

**Rule**

1. Visit root
2. Traverse left subtree
3. Traverse right subtree

**Order**

A, B, D, E, C, F

**Visualization**

```
A
├── B
│   ├── D
│   └── E
└── C
    └── F
```

Preorder “touches” the parent before children.

**Use case**

- copy tree
- serialize tree
- expression tree prefix form

---

**(b) Inorder: Left → Root → Right**

**Rule**

1. Traverse left subtree
2. Visit root
3. Traverse right subtree

**Order**
```

D, B, E, A, C, F

Use case

- for BST, inorder gives sorted order
- infix expression formatting

(c) Postorder: Left → Right → Root

Rule

1. Traverse left subtree
2. Traverse right subtree
3. Visit root

Order

D, E, B, F, C, A

Use case

- delete/free tree safely
- evaluate expression trees
- process children before parent

11.2 Breadth-First Traversal (Level Order)

Visit nodes level by level.

Order

A, B, C, D, E, F

Visualization

Level 0: A

Level 1: B, C

Level 2: D, E, F

Use case

- printing level structure
- shortest-level discovery
- insertion in complete binary tree style

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12) High-Level Architecture of Traversals in C++

12.1 Recursive DFS template

```
void traverse(Node* root) {
   if (root == nullptr) return;

   // operation
   traverse(root->left);
   traverse(root->right);
}
```

The difference between preorder, inorder, postorder is **where you put the "visit" line**.

12.2 Iterative BFS template

```
queue<Node*> q;
q.push(root);

while (!q.empty()) {
   Node* curr = q.front();
   q.pop();

   // visit curr

   if (curr->left) q.push(curr->left);
   if (curr->right) q.push(curr->right);
}
```

---

13) Operation 5 — Height of Binary Tree

Height tells how “deep” or “tall” the tree is.

13.1 Visualization

```
    1
   / </span>
  2   3
 /
4
```

Longest path from root to leaf: 1 → 2 → 4

Height = 2 (if counting edges)

13.2 Recursive Idea

Height of a node =

1 + max(height(left), height(right))

Base case:

- empty tree → -1 (if height is measured in edges) or
- empty tree → 0 (if measured in nodes)

I’ll use **height in edges** for theory clarity, but in code many use node-count-based height. I’ll point it out in code comments.

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14) Operation 6 — Count Nodes

Idea

For each node:

- count current node = 1
- count left subtree
- count right subtree

Formula:

count(root) = 1 + count(left) + count(right)

```
```

---

15) Operation 7 — Count Leaf Nodes

A node is a leaf if:

- left == nullptr
- right == nullptr

So recursively:

- if node is null → 0
- if node is leaf → 1
- else return leaf count of left + right

---

16) Operation 8 — Mirror / Invert Tree

Swap left and right child of every node.

16.1 Visualization

Original:

```
      1
     / </span>
    2   3
   / </span>
  4   5
```

Mirror:

```
      1
     / </span>
    3   2
     / </span>
    5   4
```

16.2 High-Level Architecture

At each node:

1. swap left and right
2. recursively mirror left subtree
3. recursively mirror right subtree

---

17) Complete C++ Implementation

Below is a complete C++ class implementing major binary tree operations.

17.1 Full C++ Code

```
#include <iostream>
#include <queue>
#include <algorithm>
using namespace std;

class BinaryTree {
private:
  struct Node {
    int data;
    Node* left;
    Node* right;

    Node(int value) : data(value), left(nullptr), right(nullptr) {}
  };

  Node* root;

private:
  // ---------- Recursive Helper Functions ----------

  void preorder(Node* node) {
```

```
    if (node == nullptr) return;
    cout << node->data << " ";
    preorder(node->left);
    preorder(node->right);
}

void inorder(Node* node) {
    if (node == nullptr) return;
    inorder(node->left);
    cout << node->data << " ";
    inorder(node->right);
}

void postorder(Node* node) {
    if (node == nullptr) return;
    postorder(node->left);
    postorder(node->right);
    cout << node->data << " ";
}

int height(Node* node) {
    // height in terms of edges:
    // empty tree = -1, leaf = 0
    if (node == nullptr) return -1;
    return 1 + max(height(node->left), height(node->right));
}

int countNodes(Node* node) {
    if (node == nullptr) return 0;
```

```
    return 1 + countNodes(node->left) + countNodes(node->right);
}

int countLeafNodes(Node* node) {
    if (node == nullptr) return 0;
    if (node->left == nullptr && node->right == nullptr) return 1;
    return countLeafNodes(node->left) + countLeafNodes(node->right);
}

bool searchDFS(Node* node, int key) {
    if (node == nullptr) return false;
    if (node->data == key) return true;
    return searchDFS(node->left, key) || searchDFS(node->right, key);
}

void mirror(Node* node) {
    if (node == nullptr) return;
    swap(node->left, node->right);
    mirror(node->left);
    mirror(node->right);
}

void deleteTree(Node* node) {
    if (node == nullptr) return;
    deleteTree(node->left);
    deleteTree(node->right);
    delete node;
}
```

```
public:
    BinaryTree() : root(nullptr) {}

    ~BinaryTree() {
        deleteTree(root);
    }

    // ---------- Insert using Level Order ----------
    void insert(int value) {
        Node* newNode = new Node(value);

        if (root == nullptr) {
            root = newNode;
            return;
        }

        queue<Node*> q;
        q.push(root);

        while (!q.empty()) {
            Node* current = q.front();
            q.pop();

            if (current->left == nullptr) {
                current->left = newNode;
                return;
            } else {
                q.push(current->left);
            }
```

```
      if (current->right == nullptr) {
         current->right = newNode;
         return;
      } else {
         q.push(current->right);
      }
   }
}

// ---------- Search ----------
bool search(int key) {
   return searchDFS(root, key);
}

// ---------- Traversals ----------
void preorderTraversal() {
   preorder(root);
   cout << endl;
}

void inorderTraversal() {
   inorder(root);
   cout << endl;
}

void postorderTraversal() {
   postorder(root);
   cout << endl;
```

```
}

void levelOrderTraversal() {
    if (root == nullptr) return;

    queue<Node*> q;
    q.push(root);

    while (!q.empty()) {
        Node* current = q.front();
        q.pop();

        cout << current->data << " ";

        if (current->left) q.push(current->left);
        if (current->right) q.push(current->right);
    }
    cout << endl;
}

// ---------- Height ----------
int getHeight() {
    return height(root);
}

// ---------- Count Nodes ----------
int getNodeCount() {
    return countNodes(root);
}
```

```
// ---------- Count Leaf Nodes ----------
int getLeafCount() {
    return countLeafNodes(root);
}

// ---------- Mirror Tree ----------
void invertTree() {
    mirror(root);
}

// ---------- Delete Node ----------
void deleteValue(int value) {
    if (root == nullptr) return;

    // Case: only one node
    if (root->left == nullptr && root->right == nullptr) {
        if (root->data == value) {
            delete root;
            root = nullptr;
        }
        return;
    }

    queue<Node*> q;
    q.push(root);

    Node* targetNode = nullptr;
    Node* deepestNode = nullptr;
```

```
Node* parentOfDeepest = nullptr;

while (!q.empty()) {
    deepestNode = q.front();
    q.pop();

    if (deepestNode->data == value) {
        targetNode = deepestNode;
    }

    if (deepestNode->left) {
        parentOfDeepest = deepestNode;
        q.push(deepestNode->left);
    }

    if (deepestNode->right) {
        parentOfDeepest = deepestNode;
        q.push(deepestNode->right);
    }
}

if (targetNode != nullptr) {
    // Replace target's data with deepest node's data
    targetNode->data = deepestNode->data;

    // Remove deepest node from the tree
    if (parentOfDeepest != nullptr) {
        if (parentOfDeepest->right == deepestNode) {
            parentOfDeepest->right = nullptr;
```

```
            } else if (parentOfDeepest->left == deepestNode) {
                parentOfDeepest->left = nullptr;
            }
        }

        delete deepestNode;
    }
  }
};

int main() {
  BinaryTree tree;

  // Insert nodes
  tree.insert(1);
  tree.insert(2);
  tree.insert(3);
  tree.insert(4);
  tree.insert(5);
  tree.insert(6);

  cout << "Level Order: ";
  tree.levelOrderTraversal();

  cout << "Preorder: ";
  tree.preorderTraversal();

  cout << "Inorder: ";
  tree.inorderTraversal();
```

```
    cout << "Postorder: ";
    tree.postorderTraversal();

    cout << "Search 5: " << (tree.search(5) ? "Found" : "Not Found") << endl;
    cout << "Height: " << tree.getHeight() << endl;
    cout << "Total Nodes: " << tree.getNodeCount() << endl;
    cout << "Leaf Nodes: " << tree.getLeafCount() << endl;

    tree.deleteValue(2);
    cout << "After deleting 2, Level Order: ";
    tree.levelOrderTraversal();

    tree.invertTree();
    cout << "After mirroring, Level Order: ";
    tree.levelOrderTraversal();

    return 0;
}
```

---

18) Dry Run of the Code

After insertion of: 1, 2, 3, 4, 5, 6

The tree becomes:

```
      1
     / </span>
    2   3
   /   /
  4  5 6
```

Traversals:

- Preorder: 1 2 4 5 3 6
- Inorder: 4 2 5 1 6 3
- Postorder: 4 5 2 6 3 1
- Level order: 1 2 3 4 5 6

---

19) High-Level Architecture of Each Operation (C++ Thinking)

This section is very important if you want to become strong in implementation.

19.1 Insertion Architecture

Components

- Node class/struct
- root pointer
- queue<Node*>

Flow

If root is null:

root = new node

Else:

BFS using queue

Find first missing child position

Insert there

C++ concepts involved

- dynamic memory (new)
- pointers
- STL queue
- iterative traversal

19.2 Deletion Architecture

Components

- queue for BFS

- target node pointer
- deepest node pointer
- parent-of-deepest pointer

Flow

Scan entire tree level by level

Remember:

node to delete

deepest node

Replace target data with deepest data

Disconnect deepest node

delete deepest node

C++ concepts involved

- pointer tracking
- BFS
- object lifetime
- safe pointer disconnection

19.3 Search Architecture

Recursive DFS approach

If null → false

If current matches → true

Else search left or search right

C++ concept

- recursion
- short-circuit boolean evaluation

19.4 Traversal Architecture

Preorder

visit before children

Inorder

visit between left and right

Postorder

visit after children

Level-order

use queue

19.5 Height Architecture

Recursive definition

A tree is naturally recursive:

- height of tree depends on heights of subtrees

height(node) = 1 + max(height(left), height(right))

This is why recursion feels very natural for trees.

19.6 Mirror Architecture

for every node:

swap(left, right)

recurse left

recurse right

This is a classic structural recursion problem.

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20) Time Complexity Summary

For a general binary tree

- **Insertion (level-order)**: O(n)
- **Deletion**: O(n)
- **Search**: O(n)
- **DFS Traversals**: O(n)
- **Level-order Traversal**: O(n)
- **Height**: O(n)

- **Count Nodes**: O(n)
- **Count Leaves**: O(n)
- **Mirror**: O(n)

Space

- Recursive traversals: O(h) call stack
- BFS traversals: up to O(n) queue
- where h = height of tree

---

21) Why Recursion Works So Well for Trees

A tree is recursively defined:

- a tree consists of a root
- left subtree
- right subtree

So when writing recursive tree algorithms, you usually ask:

“If I already know how to solve the problem for left subtree and right subtree, how do I combine them for the current node?”

That mindset is the key.

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22) Common Mistakes Students Make

Mistake 1: Confusing binary tree with BST

Binary tree has no ordering guarantee.

Mistake 2: Forgetting null checks

Always handle:

if (node == nullptr) return;

Mistake 3: Wrong traversal order

Preorder, inorder, postorder differ only by placement of “visit”.

Mistake 4: Memory leaks

If you allocate nodes using new, you should free memory eventually.

Mistake 5: Wrong height definition

Some define:

- empty tree = -1 Others define:
- empty tree = 0

Be consistent.

---

23) When Should You Use a Binary Tree?

Use a binary tree when:

- data is hierarchical
- recursive decomposition is natural
- you need tree traversals
- you need structured decision making
- you need variants like BST / Heap / AVL / Red-Black Tree

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24) Use Cases of Binary Tree in Backend Software Engineering

Now let's connect this to real backend engineering.

24.1 Database Indexing (via Balanced BST Concepts)

While many production databases use **B-trees/B+ trees** rather than simple binary trees, the **conceptual foundation** comes from tree-based hierarchical search structures.

Why relevant?

- fast lookup
- fast insertion
- ordered data
- range queries

Backend relevance

- indexes
- search paths
- ordered key storage

24.2 In-Memory Ordered Data Structures

Balanced binary search trees power structures like:

- ordered maps
- ordered sets
- interval search structures

In C++, std::map and std::set are usually implemented with **Red-Black Trees** (a self-balancing BST variant).

Backend use

- maintain sorted user sessions
- rank-based retrieval
- time-ordered job lookup
- range filtering

24.3 Priority Scheduling with Heaps

A **binary heap** is a complete binary tree (often stored in an array).

Use cases in backend:

- job scheduling
- task prioritization
- rate limiter queues
- message retry priority
- event processing order

Example:

- urgent emails
- highest-priority failed job processed first

24.4 Expression Trees in Query Engines / Compilers

When a backend parses:

- SQL queries
- filter expressions
- arithmetic expressions
- rules engines

it may build an **expression tree**.

Example:

```
      *
     / </span>
    +   -
   / \ / </span>
  a  b c  d
```

Represents:

(a + b) * (c - d)

Backend use

- query parsing
- rule evaluation
- execution plans
- AST-like structures

24.5 Routing / Decision Trees

Decision logic in backend can be modeled as a binary tree:

Is authenticated?

├── yes → Is admin?

│ ├── yes → admin route

│ └── no → user route

└── no → login required

Use cases

- request authorization
- fraud decision systems
- rule engines
- feature flag evaluation

24.6 Compression / Encoding (Huffman Tree)

Huffman coding uses binary trees to generate prefix codes.

Backend use

- compression
- data transmission optimization
- storage optimization

24.7 AI / ML Serving Pipelines

Decision trees are widely used in ML.

Backend use

- fraud detection API
- recommendation logic
- scoring engine
- risk classification

These are binary tree–like decision structures.

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25) Binary Tree vs Heap vs BST in Backend

This distinction is useful:

Binary Tree

General structure
No ordering guarantee

Use when:

- hierarchy matters more than ordering

Binary Search Tree

Ordered structure

Use when:

- search/insert/delete in ordered data matter

Binary Heap

Priority-based structure

Use when:

- want quick access to min/max element

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26) Interview-Oriented Understanding

If an interviewer asks:

“What is a binary tree?”

A strong answer is:

A binary tree is a hierarchical data structure in which each node has at most two children, called the left and right child. It is useful for representing recursive hierarchical relationships. Common operations include insertion, deletion, traversal, search, height calculation, and transformations like mirroring. Depending on the variant—such as BST, AVL, heap, or Red-Black Tree—it supports different performance guarantees and backend use cases.

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27) How to Study Binary Trees Properly

I recommend this sequence:

Step 1

Understand node, root, child, leaf, height

Step 2

Master recursive traversals:

- preorder
- inorder

- postorder

Step 3

Learn BFS / level order using queue

Step 4

Implement:

- insert
- search
- count
- height

Step 5

Implement:

- delete
- mirror

Step 6

Move to advanced tree variants:

- BST
- AVL
- Red-Black Tree
- Heap
- Trie
- Segment Tree

---

28) Quick Revision Sheet

Binary tree

- each node has at most 2 children

Main parts

- root
- left child
- right child

- leaf
- subtree

Traversals

- preorder = root, left, right
- inorder = left, root, right
- postorder = left, right, root
- level-order = BFS

Common operations

- insert
- delete
- search
- height
- count
- mirror

Complexity

Most operations in a general binary tree: O(n)

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29) Final Intuition Summary

A binary tree is powerful because it combines:

- **hierarchical modeling**
- **recursive structure**
- **flexible traversals**
- **foundation for many advanced structures**

If arrays are for **indexed linear data**, and linked lists are for **sequential dynamic data**, then trees are for **hierarchical recursive data**.

That is why binary trees are one of the most important topics in DSA.

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30) What You Should Do Next

To become truly comfortable, you should now practice these 5 things by hand:

1. Draw a tree and write preorder, inorder, postorder
2. Insert nodes using level-order
3. Delete a node using deepest-node replacement
4. Write recursive height/count functions
5. Trace recursion with paper

If you want, I can do **either** of these next:

Option A — Practice mode

I'll give you **10 binary tree problems with solutions** from easy to hard.

Option B — Deep implementation mode

I'll give you a **clean modern C++ version** using:

- separate Node class
- iterative + recursive versions
- better memory safety
- pretty tree printing

Option C — Interview mode

I'll teach:

- **Binary Tree vs BST vs Heap vs AVL vs Red-Black Tree** with diagrams and complexity comparison.

If you want, I can continue with **"Binary Tree Practice Problems + Answers in C++"** next.